

REVIEW



Hormone therapy in women with premature ovarian insufficiency: a systematic review and meta-analysis

**BIOGRAPHY**

Caroline R Gonçalves is an Obstetrician and Gynecologist at Hospital Sofia Feldman, Brazil. Her Masters in Women's Health at the Federal University of Minas Gerais, Brazil, involved studying premature ovarian insufficiency.

Caroline R. Gonçalves¹, Amanda S. Vasconcellos¹, Thaiana R. Rodrigues¹,
Fabio V. Comin², Fernando M. Reis^{1,*}

KEY MESSAGE

Hormone therapy with oestrogen and progesterone or progestin (including contraceptives) benefits women with premature ovarian insufficiency, not only to mitigate hypoestrogenic symptoms but also to preserve bone mineral density, avoid uterine atrophy and increase adult height. More studies are needed to confirm the long-term benefits and safety of this therapy.

ABSTRACT

The aim of this systematic review and meta-analysis was to evaluate the effectiveness of different hormone therapies, including hormonal contraceptives, in women with premature ovarian insufficiency (POI). Thirty reports of 28 studies were included, with a total of 4004 participants with POI from diverse aetiologies, of whom 3785 received hormone therapies and 219 received calcium supplementation, vitamin D, placebo or no treatment. Hormone therapy was superior to non-treatment, placebo, calcitriol or calcium in preserving bone mineral density (BMD) in women with POI. Hormone therapy was associated with up to 80% reduction in the prevalence of hot flushes and with stability or improvement in the quality of life scores. Hormone therapy induced significant increases in uterine volume and endometrial thickness in women with POI. The studies yielded convergent results and were of good quality, although some lacked blinding or had incomplete outcome data. Moderate to high quality evidence was found that hormone therapy with oestrogen and progesterone or progestin (including contraceptives) is beneficial to women with POI, not only to mitigate hypoestrogenic symptoms but also to preserve BMD and avoid uterine atrophy. More studies are needed to confirm the long-term safety of this therapy and to assess its possible impact on the risk of hard outcomes such as bone fractures and cardiovascular events.

¹ Department of Obstetrics and Gynecology, Universidade Federal de Minas Gerais, Belo Horizonte, Brazil

² Department of Internal Medicine, Universidade Federal de Minas Gerais, Belo Horizonte, Brazil

© 2022 Reproductive Healthcare Ltd. Published by Elsevier Ltd. All rights reserved.

*Corresponding author. E-mail address: fmreis@ufmg.br (F. M. Reis). <https://doi.org/10.1016/j.rbmo.2022.02.006> 1472-6483/© 2022 Reproductive Healthcare Ltd. Published by Elsevier Ltd. All rights reserved.

Declaration: The authors report no financial or commercial conflicts of interest.

KEYWORDS

Bone mineral density
Hormone therapy
Meta-analysis
Oestrogen
Premature ovarian insufficiency
Progesterone

INTRODUCTION

Premature ovarian insufficiency (POI) is an important medical issue concerning women of reproductive age. It is a clinical syndrome defined by loss of ovarian activity before the age of 40 years. Over the years, several terms have been used to describe this condition, such as 'primary ovarian insufficiency', 'premature menopause' and 'premature ovarian failure', but the latest guideline from the European Society of Human Reproduction and Embryology (ESHRE) adopted 'premature ovarian insufficiency' as the standard nomenclature (Webber *et al.*, 2016). The diagnostic criteria are not fully standardized, but do not differ that much. The ESHRE recommends the following diagnostic criteria: oligo/amenorrhoea for at least 4 months, and an elevated FSH concentration >25 IU/l on two occasions at least 4 weeks apart (Webber *et al.*, 2016). Ovarian insufficiency is caused by several different mechanisms, but in most spontaneous POI cases (90%) no cause can be found. There may be reduced peak follicle number, increased atresia of follicles through apoptosis, or failure of follicle maturation (La Marca and Mastellari, 2021; Panay *et al.*, 2020).

The prevalence of POI varies among ethnic groups across the world, and studies show a prevalence that ranges from 1% (Coulam *et al.*, 1986) up to 5.5% (Jungari and Chauhan, 2017; Luborsky *et al.*, 2003; Mishra *et al.*, 2017; Progetto Menopausa Italia Study Group, 2003). If not treated quickly, women living with POI face the short- and long-term consequences of prolonged hypoestrogenism and can develop symptoms such as lack of libido, vaginal dryness, mood disorders, rise in cardiovascular risk and cognitive impairment (Panay *et al.*, 2020). Decreased mineral density promotes osteopenia and osteoporosis, as well as bone fractures. The health provider should be aware of POI signs and symptoms in order to promote prompt diagnosis (Lambrinoudaki *et al.*, 2021). The time from onset of symptoms until diagnosis has been evaluated in only a few studies. A structured interview survey conducted in the USA found that the median time elapsed from the onset of disordered menses until the diagnosis of POI was 2 years, reaching

5 years in a quarter of the women (Alzubaidi *et al.*, 2002). A more recent case series showed that the mean time taken to diagnose POI from the onset of symptoms was 6 years (Bhatt *et al.*, 2019).

The therapeutic approach for POI patients relies mainly on hormone therapy with oestrogen and progesterone receptor ligands, which includes hormonal contraceptives. The targets for hormone therapy include improvement of cardiovascular, urogenital, bone and mental health as well as quality of life. However, most hormonal formulations, regimens and dosages prescribed to women with POI are based on studies with older women experiencing natural menopause; much less is known about the effects of different types of hormone therapy in the specific population with POI, especially when it comes to long-term results.

Therefore, the aim of this systematic review was to evaluate the available evidence about the effects of different hormone therapies on hypoestrogenic signs and symptoms, bone and cardiovascular health, and quality of life in women with POI.

MATERIALS AND METHODS

Review registration and literature search

This review followed the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) guidelines (Page *et al.*, 2021) and was registered in the International Prospective Register of Systematic Reviews (PROSPERO, CRD42018115998).

A systematic search in Allied and Complementary Medicine™, DH-DATA: Health Administration, Medical Toxicology and Environmental Health, PubMed, Embase® and MEDLINE® was performed through ProQuest, from the earliest available date to 31 May 2021 (Supplementary Figure 1). The search terms used were: (menopause hormone therapy) OR (hormone replacement therapy) OR (oestrogen replacement therapy) OR (oral contraceptive/exp) and (ovarian insufficiency) OR (ovarian failure) OR (premature ovarian failure) OR (primary ovarian insufficiency) OR (menopause, premature). Search results were imported and managed via EndNote X8 (Thomson Reuters, NY, USA).

Study selection and data extraction

Studies were included if they met the following PICOS criteria. Patients: women diagnosed with POI of any aetiology, including bilateral oophorectomy. Intervention: hormone therapy with any regimen of oestrogen, progesterone or androgen therapy, or tibolone, alone or combined (including hormonal contraceptives). Comparison: placebo, non-hormonal treatments, no treatment. Outcomes: bone mineral density (BMD), incidence of bone fractures, vasomotor symptoms (hot flushes), quality of life, cardiovascular disease, cardiovascular events, lipid concentrations, uterine morphology/blood flow, body growth, adverse effects. Study design: randomized controlled trials (RCT) or cohort studies. Only full texts in English or Portuguese were eligible.

The following exclusion criteria were used: case-control studies, cross-sectional studies, animal studies, basic experiments, literature reviews, meta-analyses, editorials, commentaries, opinion pieces, study protocols or case reports. Duplicates were removed electronically and then manually.

Two investigators (CRG and ASV) independently extracted data using a standard data collection form and any discrepancies were evaluated and discussed by a third author (FMR). The data extracted from each study included study design, year of publication, country, setting, age range of participants, criteria used to diagnose POI, aetiology of POI, number of POI cases, number of non-POI controls, drug, dose and regimen used in hormone therapy, comparator, duration of intervention (months), follow-up (months), number of participants lost to follow-up, study completion rate and main findings.

Evaluation of the methodological quality of the included studies

Two reviewers (CRG and FMR) evaluated separately the quality of the included RCT according to the Cochrane risk of bias tool for seven domains, as follows: random sequence generation (selection bias), allocation concealment (selection bias), blinding of participants and personnel (performance bias), blinding of outcome assessment (detection bias), incomplete outcome data (attrition bias), selective reporting (reporting bias) and other bias. The studies were judged as having low, unclear or high

risk of bias (Higgins and Green, 2011). Cohort studies were evaluated using the Newcastle–Ottawa Scale (Wells et al., 2000) converted to Agency for Health Research and Quality (AHRQ) standards into three categories: low, medium or high quality (Modesti et al., 2016).

Grading the quality of evidence

The quality of evidence was assessed with the Grading of Recommendations Assessment, Development, and Evaluation (GRADE) approach (Guyatt et al., 2011). For each outcome, the quality of evidence was initially classified as high (mostly based on RCT) or moderate (based on cohort studies). Two researchers then assessed downgrading factors such as serious study limitations, inconsistency of findings between studies, indirectness of outcome measures, imprecision of the results and publication bias, and upgrading factors such as large effect and dose-dependent response. Finally, the quality of evidence was graded as high, moderate, low or very low (Guyatt et al., 2011).

Meta-analysis

The only outcome with sufficient quantitative data to be meta-analysed was the variation in BMD of the lumbar spine. Data were extracted as g/cm² and the changes observed after the treatments were converted to percentage of baseline levels and adjusted for the duration of treatment (Supplementary Table 1).

The meta-analysis was conducted using RevMan 5.4 (Review Manager, 2020). A random-effects model was used to compute the pooled group differences because it was decided not to assume that all studies would share a common effect size and direction (Higgins and Green, 2011). Therefore, heterogeneity was evaluated using the τ^2 statistic, which measures the variance between studies, as indicated for random-effects analyses (Higgins and Green, 2011). The influence of a single study on the pooled effect estimate was evaluated by sensitivity analysis excluding one trial in each turn. Funnel plots were not used to assess the risk of publication bias because the number of included studies was not sufficient (Higgins and Green, 2011).

RESULTS

Literature search results

A total of 1602 records were identified through electronic database search.

After removing 411 duplicates, 1191 records were screened and 1069 were excluded, leaving 122 reports sought for retrieval. After reading titles and abstracts, 58 reports were retrieved and assessed in full text for eligibility. In this phase 28 reports were excluded for various reasons, leaving 30 reports of 28 studies to be included in the review (FIGURE 1). The included studies comprised 14 articles reporting on 12 RCT (Cartwright et al., 2016; Crofton et al., 2010; Fatemi et al., 2007; Guerrieri et al., 2014; Kraus et al., 2018; Kung et al., 1999; Langrish et al., 2009; Li et al., 1992; O'Donnell et al., 2012; Popat et al., 2014; Ross et al., 2011; Shah et al., 2014; Shea et al., 2015; Zuckerman-Levin et al., 2009) and 16 cohort studies (Achiron et al., 1995; Benetti-Pinto et al., 2015; Biljan et al., 1995; Castelo-Branco et al., 1996; Chatterjee et al., 2011; Emans et al., 1990; Gazarra et al., 2020; Gökmen and Yapar Eyi, 1999; Hartmann et al., 1997; Kalantaridou et al., 2004; Khastgir et al., 2003; Kim et al., 2016; Naessén et al., 2014; Nakamura et al., 2015; Vermeulen et al., 2017; Yang et al., 2017).

Characteristics of participants

The studies included a total of 4004 participants with POI (median 31, range 6 to 2184 participants per study), of whom 3785 received hormone therapy and 219 received calcium supplementation, vitamin D, placebo or no treatment (TABLE 1). The mean age of participants at the time of enrolment varied according to the predominant aetiology of POI, being typically younger in studies of patients with Turner syndrome only (Khastgir et al., 2003; Kraus et al., 2018; Nakamura et al., 2015; Ross et al., 2011; Zuckerman-Levin et al., 2009) than in patients with bilateral oophorectomy only (Gökmen and Yapar Eyi, 1999; Vermeulen et al., 2017). Most participants were selected and followed in hospitals, and the geographical distribution included North and South America, Europe and Asia (TABLE 1).

Interventions

Oestrogen types included conjugated equine oestrogens, oestradiol valerate and ethinyl oestradiol by oral route, and 17 β oestradiol by transdermal route, either continuous or 21 days per month. Ethinyl oestradiol was used only in combined oral contraceptives (COC). Progestins were administered in cyclic

regimens during 10 to 21 days per month, while tibolone was used continuously (TABLE 1).

Outcomes

The selected studies evaluated most of the outcomes pre-specified in the review protocol (TABLE 1). However, there were two exceptions, as the literature search did not return any RCT or cohort study evaluating the effects of hormone therapy on the incidence of bone fractures or cardiovascular events (myocardial infarction, thromboembolism, stroke) in women with POI (TABLE 2).

Excluded studies

Most studies that were about normal menopause or did not evaluate the effects of hormone therapy were excluded at the screening phase. Among the 28 articles excluded through full text reading, 13 were excluded because the study participants were on average older than 40 years at baseline (Baksu et al., 2007; Basbug et al., 1997; Bhattacharya and Jha, 2010; Castelo-Branco et al., 1993; Eeles et al., 1991; Haines et al., 1995; Kilic et al., 2010; Lip et al., 1997; Lyritis et al., 1995; Madalinska et al., 2006; Sherwin and Gelfand, 1985a,b; Somunkiran et al., 2007), and seven because the study interventions (Bakhsh et al., 2015; Christ et al., 2018; Honigberg et al., 2019; Wong et al., 2018) or outcomes (Elkind-Hirsch et al., 1992, 1993; Kapetanakis et al., 1982) were not compatible with the review entry criteria (FIGURE 1). Five studies were also excluded that evaluated hormone therapy in women with POI but had a cross-sectional design (Bachelot et al., 2016; Benetti-Pinto et al., 2019; Cardona Attard et al., 2019; Giraldo et al., 2017; Kurabayashi et al., 1993), one study protocol (Mittal et al., 2013), one possible overlap with another report (Li et al., 1991) and a review (Atsma et al., 2006).

Summary of results

All studies have consistently shown that hormone therapy in different regimens increased or conserved BMD. Five studies (three RCT) showed that hormone therapies were superior to non-treatment, placebo, calcitriol or calcium in preserving BMD in women with POI. As shown in FIGURE 2A, meta-analysis of RCT indicated that hormone therapy induced a greater gain in BMD compared with control (mean difference 3.19% BMD increase per year, 95% CI 1.84–4.53). When the analysis excluded

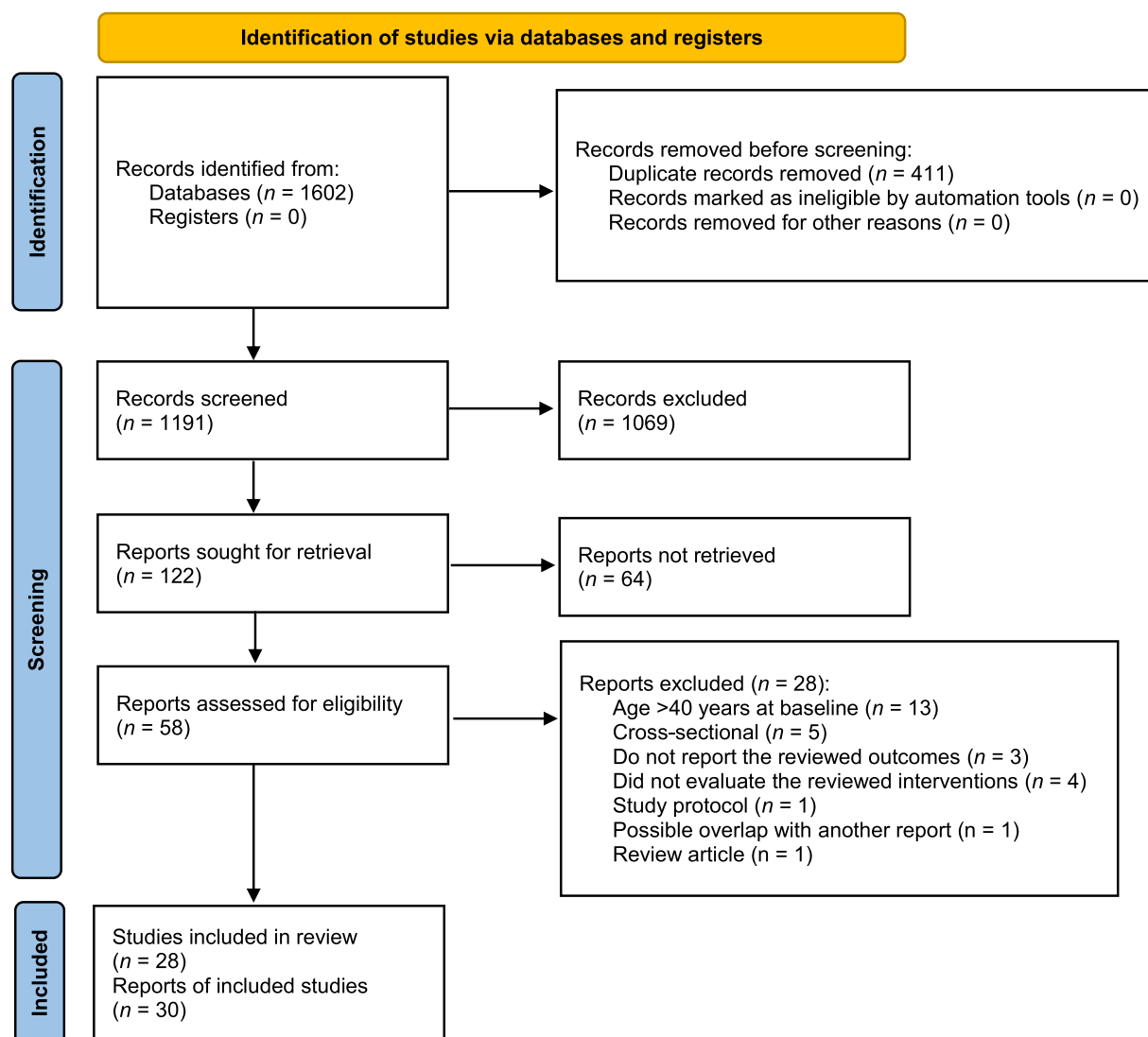


FIGURE 1 Study flow chart showing the selection of studies for inclusion. The 30 reports included in the review concerned 28 studies because one study generated three reports (Crofton *et al.*, 2010; Langrish *et al.*, 2009; O'Donnell *et al.*, 2012).

COC and considered only formulations labelled for menopausal hormone therapy (FIGURE 2B), the BMD gain remained greater in the hormone therapy group than in the control group (mean difference 3.55% BMD increase per year, 95% CI 2.08–5.02). The same extent of hormone therapy benefit was seen when adding a cohort study to the pooled data for treatment with formulations labelled for menopausal hormone therapy only (3.27% BMD increase per year, 95% CI 1.89–4.65, FIGURE 2C). There was a significant difference between cyclic COC, with 7-day hormone-free intervals each month, and continuous menopausal hormone therapy as regards BMD gain (1.95% BMD increase per year, 95% CI 0.48–3.43, FIGURE 3A). The design of both studies included in this comparison

did not allow us to distinguish between the effects of different formulations and different regimens (Cartwright *et al.*, 2016; Crofton *et al.*, 2010). However, a recent cohort study showed that BMD at the lumbar spine and total femur increased more in patients receiving continuous COC or high-dose conjugated oestrogens than in women receiving standard dose menopausal hormone therapy (Gazarra *et al.*, 2020). Also, adding testosterone to the oestrogen–progesterone regimen did not change BMD gain (FIGURE 3B).

Vasomotor symptoms were measured in two cohort studies. Hormone therapy was associated with up to 80% reduction in the prevalence of hot flushes (Vermeulen *et al.*, 2017) and 50%

reduction in Kupperman menopausal index (Yang *et al.*, 2017). Two RCT evaluated quality of life (Guerrieri *et al.*, 2014; Zuckerman-Levin *et al.*, 2009). Although designed to compare groups treated with and without testosterone, both studies showed that women treated with oestrogen plus progestin had stability or improvement in the quality of life scores after 1 year.

Cardiovascular health was assessed in one small RCT whose end-points were blood pressure, arterial stiffness, urea, creatinine, renin and angiotensin II (Langrish *et al.*, 2009). In this study, hormone therapy with transdermal oestradiol and vaginal progesterone resulted in lower blood pressure compared with ethinyl oestradiol and norethisterone. Circulating lipids

TABLE 1 SUMMARY OF THE STUDIES INCLUDED IN THE SYSTEMATIC REVIEW

Study	Design	Cases	Age (years) ^a	Aetiology	Place	Setting	Treatment	COC included	No. treated	Control / comparator	No. of controls	Intervention (months)	Follow-up (months)	Lost to follow-up	Outcomes
Achiron 1995	Cohort	18	-	Not specified	Israel	Hospital	Oral oestradiol valerate, variable doses + vaginal progesterone 100 mg/d for 6 d	No	18	Baseline measures	-	1	1	0	Resistance index in endometrial blood flow measured by transvaginal colour Doppler
Biljan 1995	Cohort	16	31 ± 3	Turner syndrome, ovarian surgery, idiopathic	England, UK	Hospital	Oestradiol valerate 2 mg oral + norgestrel 500 µg 10 d/28 d oral	No	16	Baseline measures	-	0.5	0.5	0	Pulsatility index of the uterine arteries
Castello-Branco 1996	Cohort	13	30 (19–39)	Bone marrow transplantation	Spain	Hospital	CEE 0.625 mg or 17β-oestradiol 50 µg + MPA 5 mg 12 d/month	No	13	Baseline measures	-	12	12	0	BMD, liver enzymes
Chatterjee 2011	Cohort	375	34 ± 3	Not specified	England, UK	Hospital	CEE 0.625 mg + MPA 10 mg	No	375	Baseline measures	-	>12	60	Unknown	BMD, bone turnover markers
Emans 1990	Cohort	28	20 ± 3	Turner syndrome, radiation therapy, idiopathic	USA	Hospital	CEE 0.625 mg 21 d/month + MPA 10 mg/12 d/month oral	No	21	No treatment	7	24	24	0	BMD
Fatemi 2007	RCT crossover	6	37 ± 10	Not specified	Belgium	Hospital	Oestradiol valerate 4–6 mg/26 d/month + dydrogesterone 20 mg oral 12 d/month	No	6	Oestradiol valerate 4–6 mg/26 d/month + progesterone 200 mg vaginal 12 d/month	6 ^d	1	1	0	Endometrial histology
Gökmen 1999	Cohort	2184	45 ± 4 ^b	Bilateral oophorectomy	Turkey	Hospital	CEE 0.625 mg/30 d/month oral	No	1102	Transdermal oestradiol 50 µg/d	1082	12–84	84	0	Lipid-lipoprotein concentrations
Guerrieri 2014	RCT	128	32 ± 6	Spontaneous	USA	Hospital	Oestradiol 100 µg transdermal + MPA 10 mg 12 d/28 d oral + testosterone 150 µg transdermal	No	67	Oestradiol 100 µg transdermal + MPA 10 mg 12d/28d oral + placebo	61	12	12	5	Quality of life, self-esteem, mood
Hartmann 1997	Cohort	24	29 ± 4	Idiopathic	Austria	Hospital	Oestradiol valerate 2 mg + progesterone 400 mg 10 d/month	No	24	Baseline measures	-	12	12	0	Plasma growth hormone and insulin-like growth factor 1 concentrations post-gonadotrophin-releasing hormone injection
Kalanitidou 2004	Cohort	18	35 ± 5	Not specified	Greece	Hospital	CEE 0.625 mg/30 d/month oral + MPA 5 mg cyclically	No	18	Baseline measures	-	6	6	0	Endothelial function, lipids
Khasigir 2003	Cohort	21	31 (20–40)	Turner syndrome	England, UK	Hospital	Oestradiol 50 mg subcutaneous implant + MPA 5 mg 10 d/month oral	No	21	Baseline measures	-	36	36	0	BMD

(continued on next page)

TABLE 1 (continued)

Study	Design	Cases	Age (years) ^a	Aetiology	Place	Setting	Treatment	COC included	No. treated	Control / comparator	No. of controls	Intervention (months)	Follow-up (months)	Lost to follow-up	Outcomes
Kim 2016	Cohort	25	15 (11–19)	Turner syndrome, chemotherapy	Korea	Medical centre	Oestradiol valerate 0.5–2 mg + MPA 10 mg/12 d/month oral	No	25	Baseline measures	–	24	24	0	Ultrasonographic uterine changes
Kraus 2018	RCT	11	15 ± 7	Turner syndrome	Chile	Hospital	17β oestradiol 0.5 mg/21 d/month (titration to 1, 1.5, 2 mg according to serum oestradiol concentrations) oral + MPA 10 mg/7 d/month oral	No	7	Transdermal oestradiol 21 d/month + MPA 10 mg 7 d/month oral	4	12	12	0	Ultrasonographic uterine changes
Kung 1999	RCT	28	36 ± 6	Systemic lupus erythematosus	Hong Kong, China	Hospital	CEE 0.625 mg/21 d/month oral + MPA 5 mg/12 d/month oral + calcium carbonate 1 g/30 d/month oral	No	13	Calcitriol + calcium	15	24	24	0	Prevention of bone loss, systemic lupus erythematosus disease activity
Li 1992	RCT crossover	21	33 (27–38)	Turner syndrome, chemotherapy, radiation therapy, idiopathic	England, UK	Hospital	Oestradiol valerate 1, 2 or 4 mg 21 d/month oral + norgestrel 500 mg 10 d/month	No	21	Oestradiol valerate at variable doses	21 ^d	1	2	0	Endometrial thickness, endometrial morphology and mitoses
Nakamura 2015	Cohort	88	26 ± 7	Turner syndrome	Japan	Hospital	CEE 0.625 mg/21 d/month oral + MPA 5 mg/10 d/month	No	72	No treatment	16	60–384	60–384	0	Uterine growth, BMD
Popat 2014	RCT	145	32 ± 6	Spontaneous with normal karyotype	USA	Community	Oestradiol 100 µg 30 d/month transdermal + MPA 10 mg/12 d/month oral + calcium + testosterone patch 150 µg 30 d/month transdermal	No	73	Oestradiol 100 µg 3072 d/month transdermal + MPA 10 mg/12 d/month oral + calcium + placebo	36	36	10	BMD	
Ross 2011	RCT	137	9 ± 3	Turner syndrome	USA	Primary care	Oral oestradiol + growth hormone (oestradiol + placebo)	No	33 (37)	Placebo + growth hormone (double placebo)	34 (33)	132	132	2	Height
Shah 2014	RCT	20	14 ± 1	Turner syndrome, bone marrow transplantation	USA	Hospital	CEE, oral 17β-oestradiol or transdermal 17β-oestradiol + micronized progesterone 200 mg/10 d/month	No	20	Baseline measures	–	24	24	3	Height, feminization, puberty, biochemical markers
Shea 2015	RCT	70	36 ± 8	Leuprolide acetate for 20 weeks	USA	Hospital	Oestradiol 75 µg continuous	No	35	Placebo	35	5	5	9	Body composition (fat-free mass), BMD

(continued on next page)

TABLE 1 (continued)

Study	Design	Cases	Age (years) ^a	Aetiology	Place	Setting	Treatment	COC included	No. treated	Control / comparator	No. of controls	Intervention (months)	Follow-up (months)	Lost to follow-up	Outcomes
Vermeulen 2017	Cohort	57	39 ± 4	Risk-reducing bilateral salpingo-oophorectomy	Netherlands	Hospital	Tibolone or oestrogen and progestin (oral, transdermal or topical)	No	27	No treatment	30	9	9	4	Vasomotor symptoms, sexual functioning
Yang 2017	Cohort	130	29 ± 9	Chemotherapy	China	Hospital	Oestradiol valerate 2 mg/d 21 d/month + dydrogesterone 10 mg or cyproterone acetate 1 mg 10 d/month	No	130	Baseline measures	–	24	24	17	Kupperman menopausal index, vasomotor symptoms
Benetti-Pinto 2015	Cohort	72	34 ± 7	Not specified	Brazil	Hospital	CEE + MPA, 17β-oestradiol + norethisterone, or EE + levonorgestrel	Yes	72	Baseline measures	–	>24	24–96	0 (24 months); 54 (96 months)	BMD
Cartwright 2016	RCT	59	40 (35–43) (IQR)	Spontaneous	England, UK	Hospital	Oestradiol 2 mg 30 d/month oral + levonorgestrel 75 µg 12 d/month oral, or EE 30 µg + levonorgestrel 150 µg 21 d/month oral	Yes	30	No treatment	29	24	24	23	BMD, bone turnover markers
Crofton 2010 ^c	RCT crossover	34	27 (23–30) (95% CI)	Turner syndrome, chemotherapy, radiation therapy, oophorectomy, idiopathic	Scotland, UK	Hospital	EE 30 µg 21 d/month oral + norethisterone 1.5 mg/21 d/month oral	Yes	34	Oestradiol 100 µg 7 d/month transdermal + oestradiol 150 µg 21 d/month transdermal + progesterone 400 mg 14 d/month vaginal	34 ^d	12	12	16	BMD, bone turnover markers
Gazarra 2020	Cohort	210	30 ± 9	Not specified	Brazil	Hospital	EE 30 µg + levonorgestrel, continuously, or CEE 0.625–1.25 mg + oral progestin, or tibolone 2.5 mg daily	Yes	190	No treatment	20	24	24	0	BMD
Langrish 2009 ^c	RCT crossover	34	25 (19–39)	Turner syndrome, chemotherapy, radiation therapy, oophorectomy, idiopathic	Scotland, UK	Hospital	EE 30 µg 21 d/month oral + norethisterone 1.5 mg/21 d/month oral	Yes	34	Oestradiol 100 µg 7 d/month transdermal + oestradiol 150 µg 21 d/month transdermal + progesterone 400 mg 14 d/month vaginal	34 ^d	12	24	12	Cardiovascular health: blood pressure, serum angiotensin concentrations, plasma creatinine concentrations
Naessén 2014	Cohort	51	27 (15–43)	Chemotherapy	Sweden	Community	Combined oral contraceptive	Yes	51	Baseline measures	–	108–156	108–156	3	BMD, reproductive function, serum PTH and calcium concentrations

(continued on next page)

TABLE 1 (continued)

Study	Design	Cases	Age (years) ^a	Aetiology	Place	Setting	Treatment	COC included	No. treated	Control / comparator	No. of controls	Intervention (months)	Follow-up (months)	Lost to follow-up	Outcomes
O'Donnell 2012 ^c	RCT crossover	34	27 (19–40)	Turner syndrome, chemo-therapy, radiation therapy, oophorectomy, idiopathic	Scotland, UK	Hospital	EE 30 µg 21 d/month oral + norethisterone 1.5 mg/21 d/month oral	Yes	34	Oestradiol 100 µg 7 d/month transdermal + oestradiol 150 µg 21 d/month transdermal + progesterone 200 mg 2xd 14 d/month vaginal	34 ^d	12	12	4	Uterine volume, endometrial thickness, blood flow
Zuckerman-Levin 2009	RCT crossover	15	21 ± 3	Turner syndrome	Israel	Hospital	EE 20 µg + gestodene 75 µg 21 d/month oral + methyl testosterone 1.5 mg/30 d/month oral	Yes	15	EE 20 µg + gestodene 75 µg 21 d/month oral	15 ^d	12	24	1	Body composition, BMD, lipids, neurocognitive evaluation, quality of life

BMD = bone mineral density; CEE = conjugated equine oestrogens; d = days; EE = ethinyl oestradiol; IQR = interquartile range; MPA = medroxyprogesterone acetate; RCT = randomized controlled trial.

^a Mean ± SD or median (range) unless otherwise stated.^b 8.6 ± 6.9 years post-oophorectomy.^c Same RCT.^d Same participants who received the other treatment in the crossover design.

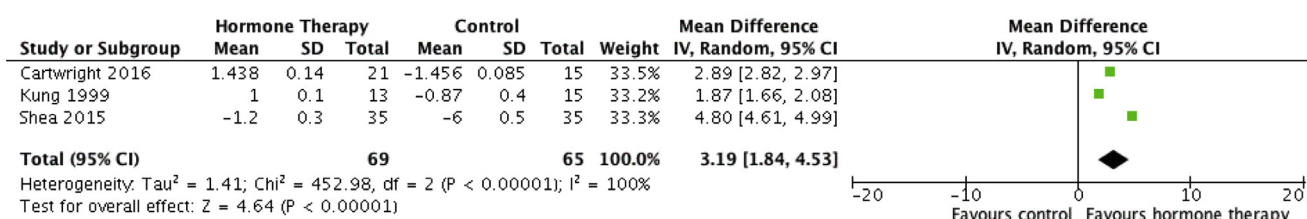
TABLE 2 GRADING OF RECOMMENDATIONS ASSESSMENT, DEVELOPMENT, AND EVALUATION (GRADE) SYNTHESIS OF THE EVIDENCE

Outcome	Studies and participants	Limitations	Inconsistency	Indirectness	Imprecision	Publication bias	Findings	Quality of evidence
Bone mineral density	6 RCT and 8 cohort studies, 1209 participants	Serious. Most RCT had high risk of bias.	No concerns	No concerns	No concerns	No concerns	All studies have consistently shown that HT in different regimens increased or conserved BMD. Five studies (3 RCT) showed that HT were superior to non-treatment, placebo, calcitriol or calcium to preserve BMD in women with POI. Two studies evaluated adding testosterone on the oestrogen-progesterone regimen and there was no difference in BMD.	⊕⊕⊕⊕ Moderate
Bone fractures	x	x	x	x	x	x	No studies have evaluated the incidence of bone fractures.	x
Vasomotor symptoms	2 cohort studies, 187 participants	No concerns	No concerns	No concerns	No concerns	No concerns	HT was associated with up to 80% reduction in the prevalence of hot flushes and 50% reduction in Kupperman menopausal index.	⊕⊕⊕⊕ Moderate
Quality of life	2 RCT, 143 participants	Serious. There were no studies comparing HT vs no therapy or placebo. Both RCT compared EP vs EPT.	No concerns	No concerns	No concerns	No concerns	Both studies showed that women treated with EP therapy had stability or improvement in the quality of life scores after 1 year.	⊕⊕⊕⊕ Moderate
Cardiovascular disease	1 RCT, 34 participants	Serious. The study was open-label and there was no control (untreated) group.	Does not apply	Serious. The outcomes were blood pressure, arterial stiffness, urea, creatinine, renin and angiotensin II.	No concerns	No concerns	HT with transdermal oestradiol and vaginal progesterone resulted in lower blood pressure compared to ethinyl estradiol and norethisterone.	⊕⊕⊕⊕ Low
Cardiovascular events	x	x	x	x	x	x	No studies have evaluated the incidence of cardiovascular events.	x
Lipids	1 RCT and 2 cohort studies, 2217 participants	In the largest cohort study the follow-up extends beyond age 40, although the mean age at oophorectomy was 36 years.	No concerns	No concerns	No concerns	No concerns	A small RCT (n = 15) compared HT regimens with and without testosterone. Total cholesterol, HDL and triglyceride concentrations were unchanged in the EP only group and decreased in the EPT group. A cohort study (n = 2184) showed a reduction of total cholesterol and LDL and a rise in HDL with HT. Triglycerides decreased with transdermal oestradiol (with or without progesterone). CEE groups had a rise in triglycerides.	⊕⊕⊕⊕ High
Uterine morphology and blood flow	4 RCT and 4 cohort studies, 219 participants	The RCT had no placebo or untreated groups.	No concerns	No concerns	No concerns	No concerns	HT induced significant increases in uterine volume and endometrial thickness. Uterine volume increased regardless of oestrogen type or route. Endometrial blood flow increased during HT accompanying plasma oestradiol concentrations.	⊕⊕⊕⊕ Moderate
Body growth	2 RCT, 157 participants	No concerns	No concerns	No concerns	No concerns	No concerns	In prepubertal girls with Turner's syndrome, the addition of oestrogen to GH increased the adult height compared to GH alone. In adolescent girls with hypogonadism, height increased similarly among different schemes of HT.	⊕⊕⊕⊕ High
Adverse effects	2 RCT and 1 cohort study, 62 participants	Serious. The studies were not powered to this outcome	No concerns	No concerns	No concerns	No concerns	After HT there was no change in liver enzymes, endometrial mitoses or lupus erythematosus disease activity.	⊕⊕⊕⊕ Moderate

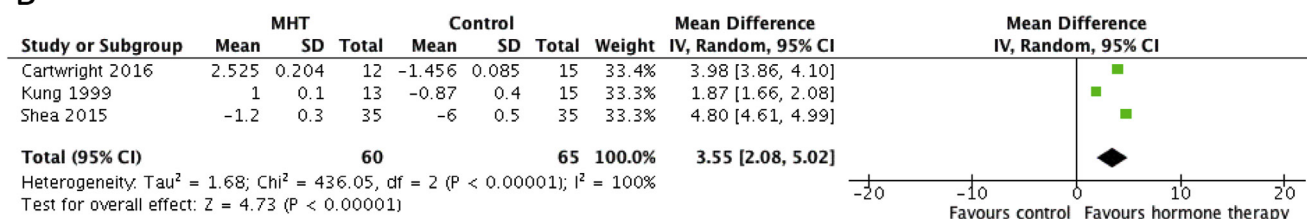
BMD = bone mineral density; CEE = conjugated equine oestrogens; EP = oestrogen-progesterin; EPT = oestrogen-progesterin-testosterone; GH = growth hormone; HDL = high-density lipoprotein cholesterol; HT = hormone therapy;

LDL = low-density lipoprotein cholesterol; POI = premature ovarian insufficiency; RCT = randomized controlled trial.

A



B



C

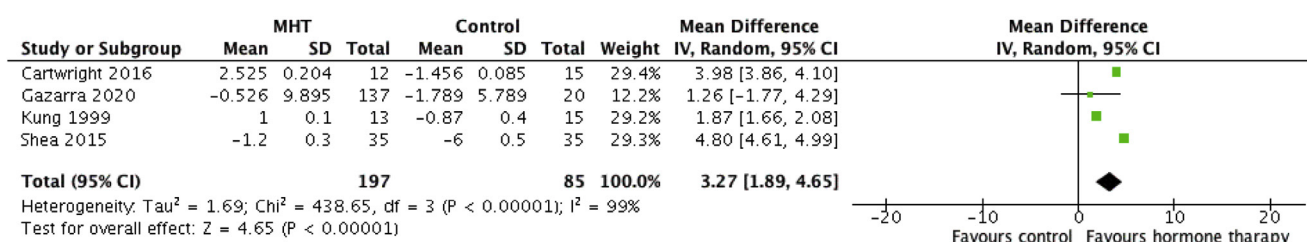


FIGURE 2 Effects of hormone therapy on bone mineral density in women with premature ovarian insufficiency (POI). Data are shown as mean difference [95% confidence interval] between hormone therapy and control groups. Data were extracted as g/cm^2 and the changes observed after the treatments were converted to percentage of baseline concentrations and adjusted for the duration of treatment. (A) Only randomized controlled trials (RCT), all types of hormone therapy. (B) Only RCT, excluding treatment with combined oral contraceptives (COC). (C) RCT and a cohort study, excluding treatment with COC.

and lipoproteins were measured in one RCT (Zuckerman-Levin *et al.*, 2009) and two cohort studies (Gökmen and Yapar Eyi, 1999; Kalantaridou *et al.*, 2004). The largest cohort (Gökmen and Yapar Eyi, 1999) showed a reduction of total cholesterol and LDL and a rise in HDL concentrations during hormone therapy, compared with baseline. Triglycerides decreased with transdermal oestradiol (with or without progesterone), whereas women using conjugated equine oestrogens had a rise in plasma triglycerides (Gökmen and Yapar Eyi, 1999).

Hormone therapy induced significant increases in uterine volume and endometrial thickness in women with POI (Kim *et al.*, 2016; Kraus *et al.*, 2018; O'Donnell *et al.*, 2012). Uterine volume increased regardless of oestrogen type or

route. Endometrial blood flow increased during hormone therapy accompanying plasma oestradiol concentrations (Achiron *et al.*, 1995; Biljan *et al.*, 1995), while endometrial histology acquired a secretory pattern in women treated with oestradiol followed by micronized progesterone or dydrogesterone (Fatemi *et al.*, 2007).

In prepubertal girls with Turner's syndrome, the addition of oestrogen increased the adult height compared with growth hormone alone (Ross *et al.*, 2011). In adolescent girls with hypogonadism, height increased similarly among different schemes of hormone therapy (Shah *et al.*, 2014).

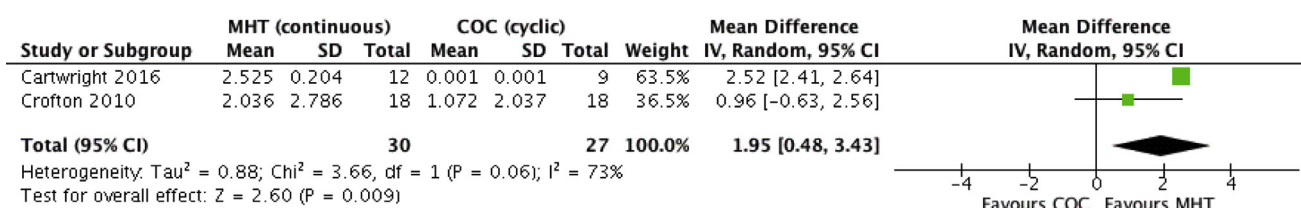
Adverse effects of hormone therapy were evaluated in three of the selected studies, with a total of only 62 participants. In

these studies, hormone therapy was not associated with significant changes in liver enzymes (Castelo-Branco *et al.*, 1996), endometrial mitoses (Li *et al.*, 1992) or lupus erythematosus disease activity (Kung *et al.*, 1999).

Heterogeneity

In the meta-analysis, the maximum between-study variance (τ^2) and SD (τ) were 1.69 and 1.30, respectively, suggesting that the effect sizes varied between studies less than the pooled mean difference between groups. Moreover, all included studies had the same direction of effect with hormone therapy being associated with higher BMD gain compared with placebo (FIGURE 2). Sensitivity analysis with the exclusion of study by study showed no significant modification in the results (data not shown).

A



B

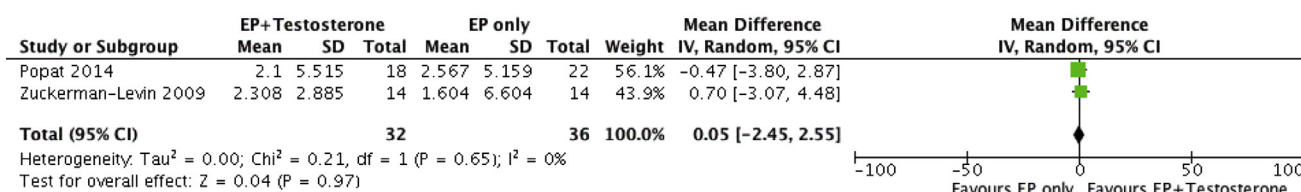


FIGURE 3 Effects of different types of hormone therapy on bone mineral density in women with premature ovarian insufficiency (POI). Data are shown as mean difference [95% confidence interval] between groups. Data were extracted as g/cm^2 and the changes observed after the treatments were converted to percentage of baseline concentrations and adjusted for the duration of treatment. (A) Comparison between continuous 'menopausal' hormone therapies (MHT) and cyclic combined oral contraceptives (COC). (B) Comparison between oestrogen and progestin (EP) plus testosterone and EP alone.

Risk of bias

Among the included RCT, there was no evidence of selection bias, although only half of the reports included details about the method of randomization (FIGURE 4). The most concerning risk was that of performance bias due to open label rather than blind interventions. However, blinding of outcome assessment was often reported even in studies that opened the intervention to the participants (Crofton *et al.*, 2010; Fatemi *et al.*, 2007; Kung *et al.*, 1999; Langrish *et al.*, 2009), therefore the risk of detection bias was high in only one RCT (Cartwright *et al.*, 2016). Most included RCT had no evidence of attrition or reporting biases (FIGURE 4 and Supplementary Figure 2).

The included cohort studies were all classified as having good quality after assessment of selection, comparability and outcome domains of the Newcastle-Ottawa Scale (Supplementary Table 2).

Quality of the evidence

As shown in TABLE 2, only low-quality evidence about the effects of hormone therapy on cardiovascular disease markers in women with POI was found. Moderate-quality evidence was found about the effects of hormone therapy on BMD, vasomotor symptoms, quality of life, uterine morphology/blood flow

and adverse effects, whereas high quality evidence was available regarding lipids and body growth (TABLE 2).

DISCUSSION

This study aimed to explore the association between the use of hormone therapy and several clinical outcomes in POI patients. Available data made possible a meta-analysis to stipulate BMD changes with and without hormone therapy. The results confirm that hormone therapy with oestrogen and progesterone or progestin either increases or stabilizes BMD. There was no improvement in BMD when testosterone was added to the scheme. Hormone therapy use was associated with several benefits, such as reduction in Kupperman menopausal index, total cholesterol, LDL and the prevalence of hot flushes. It has also increased or preserved quality of life scores and improved endometrial blood flow. None of the studies included in the systematic review evaluated frequency of bone fractures or cardiovascular events (stroke or myocardial infarction), whereas adverse effects were seldom evaluated.

Those findings support the indication of hormone therapy with oestrogen and progestogen for POI patients, bringing to light specific benefits of ovarian steroid

replacement in this group. This study helps providers to align the expectations towards the medical management of POI and supports decision-making in clinical practice.

A systematic review published in 2017 by Burgos *et al.* found that in women with POI the use of hormone therapy either maintains lipid levels or associates with a reduction in LDL and an increase in triglycerides. As for bone health, oestrogen + progestogen therapy either stabilizes or increases bone mass, compared with non-POI controls. Regarding quality of life, the review included only one study that reported an important reduction in vasomotor symptoms after the therapy was established (Burgos *et al.*, 2017).

In this systematic review, we made, for the first time in the literature, a meta-analysis evaluating changes in BMD and the use of menopausal hormone therapy or COC. The results showed that hormone therapy led to a greater gain in BMD compared with controls. We identified a possible benefit regarding continuous versus cyclic hormone therapy in relation to bone mass gain; however, this was based on RCT that used menopausal hormone therapy in continuous administration and COC in cyclic regimen (Cartwright *et al.*,

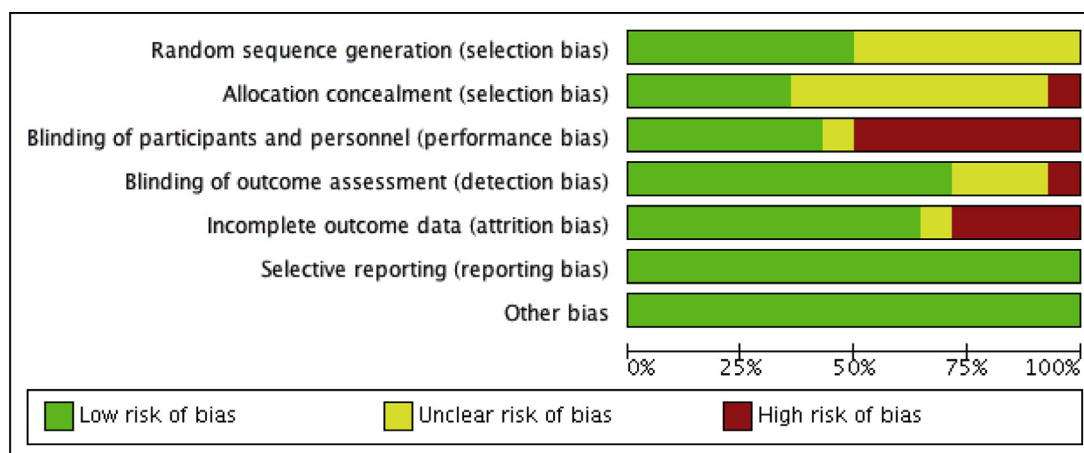


FIGURE 4 Results of the risk of bias assessment.

2016; Crofton *et al.*, 2010), making it impossible to distinguish the effects of different formulations from those of different therapeutic schedules. This can be a basis for further investigations about which hormonal formulations are most suitable for women with POI. Essentially, the BMD data are reassuring for all regimens evaluated and the choice may consider other aspects such as the need of contraception (considering the rare but existing probability of natural conception in some subgroups of POI (Dragojević Dikić *et al.*, 2020)) and patient preferences and adherence. In this context, the use of COC in continuous regimen appears to be a safe and effective option to preserve bone health (Gazarra *et al.*, 2020).

As for cardiovascular health, the findings indicate different outcomes in the lipid profile depending on the hormone route of administration, with a better plasma triglyceride profile with the use of transdermal oestradiol. Furthermore, another benefit of transdermal and vaginal hormone administration was the lower blood pressure profile. Findings about the benefits of hormone therapy on vasomotor symptoms in women with POI are consistent with previous reviews (Burgos *et al.*, 2017; Webber *et al.*, 2017). We also found relevant information about the assessment of the Kupperman menopausal index, quality of life scores and height in adolescents, in addition to measurements of endometrial thickness, uterine volume and endometrial blood flow. Scarce information was obtained about adverse effects, and all three studies that documented such effects were published before the year 2000

(Castelo-Branco *et al.*, 1996; Kung *et al.*, 1999; Li *et al.*, 1992).

This review has some strengths such as the comprehensive search of relevant clinical outcomes and the investigation of possible differences between hormone therapy drugs and regimens, which have practical importance for decision-making. The included observational studies were of good quality and it was possible to perform a meta-analysis for BMD. There are also several limitations, some of them related to included RCT that lacked blinding of participants and personnel or had incomplete outcome data. The number of subjects of the meta-analysis was small, although the findings were predominantly consistent between studies. The origin of the participants was from teaching hospital clinics, making it difficult to associate the findings with the general population, but this is not a problem to estimate the effect of treatments. The study could not provide a simple answer to which is the best hormone therapy regimen for women with POI, although we found evidence that any hormone therapy is better than no hormone therapy, and that non-oral administration results in lower blood pressure and triglycerides.

In summary, this systematic review found moderate to high quality evidence that hormone therapy with oestrogen and progesterone or progestin (including contraceptives) is beneficial to women with POI, not only to mitigate hypoestrogenic symptoms but also to preserve BMD, reduce plasma cholesterol concentrations, avoid uterine atrophy and increase adult height (in prepubertal girls with Turner's syndrome).

More studies are needed to reassure the long-term safety of this therapy and to assess its possible impact on the risk of hard outcomes such as bone fractures and cardiovascular events.

ACKNOWLEDGEMENTS

We are grateful to Professor Cristina Laguna Benetti-Pinto and Dr Andrea Nácul for their critical reading and helpful suggestions to this review.

This study was supported by the Brazilian National Institute of Hormones and Women's Health/Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq INCT 465482/2014-7).

SUPPLEMENTARY MATERIALS

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.rbmo.2022.02.006](https://doi.org/10.1016/j.rbmo.2022.02.006).

REFERENCES

- Achiron, R., Levran, D., Sivan, E., Lipitz, S., Dor, J., Mashiach, S. **Endometrial blood flow response to hormone replacement therapy in women with premature ovarian failure: a transvaginal Doppler study.** *Fertility and Sterility* 1995; 63: 550–554
- Alzubaidi, N.H., Chapin, H.L., Vanderhoof, V.H., Calis, K.A., Nelson, L.M. **Meeting the needs of young women with secondary amenorrhea and spontaneous premature ovarian failure.** *Obstetrics and Gynecology* 2002; 99: 720–725
- Atsma, F., Bartelink, M.-L.E.L., Grobbee, D.E., van der Schouw, Y.T. **Postmenopausal status and early menopause as independent risk factors for cardiovascular disease: a meta-analysis.** *Menopause (New York, N.Y.)* 2006; 13: 265–279
- Bachelot, A., Nicolas, C., Gricourt, S., Dulon, J., Lebar, M., Golmard, J.L., Touraine, P. **Poor compliance to hormone therapy and decreased bone mineral density in women with premature ovarian insufficiency.** *PloS One* 2016; 11e0164638
- Bakhsh, H., Dei, M., Bucciantini, S., Balzi, D., Bruni, V. **Premature ovarian insufficiency in young girls: repercussions on uterine volume and bone mineral density.** *Gynecological Endocrinology* 2015; 31: 65–69
- Baksu, B., Davas, I., Agar, E., Akyol, A., Uluocak, A. **Do different delivery systems of estrogen therapy influence serum lipids differently in surgically menopausal women?** *The Journal of Obstetrics and Gynaecology Research* 2007; 33: 346–352
- Basbug, M., Aygen, E., Tayyar, M., Muhtaroglu, S., Demir, I., Okten, S. **Twenty two weeks of transdermal estradiol increases sex hormone-binding globulin in surgical menopausal women.** *European Journal of Obstetrics, Gynecology, and Reproductive Biology* 1997; 73: 149–152
- Benetti-Pinto, C.L., Ferreira, V.B., Yela, D.A. **Long-term follow-up of bone density in women with primary ovarian insufficiency.** *Menopause (New York, N.Y.)* 2015; 22: 946–949
- Benetti-Pinto, C.L., Menezes, C., Yela, D.A., Cardoso, T.M. **Sleep quality and fatigue in women with premature ovarian insufficiency receiving hormone therapy: a comparative study.** *Menopause (New York, N.Y.)* 2019; 26: 1141–1145
- Bhatt, S.J., O'Besso, V.S., Douglas, N.C., McGovern, P., Loughlin, J., Morelli, S.S. **Prolonged time to diagnosis of primary ovarian insufficiency (POI) in an urban reproductive endocrinology (RE) clinic.** *Fertility and Sterility* 2019; 112: e354
- Bhattacharya, S.M., Jha, A. **Effects of transdermal estradiol gel and oral tibolone on health-related quality of life after surgical menopause.** *International Journal of Gynaecology and Obstetrics* 2010; 110: 213–216
- Biljan, M.M., Taylor, C.T., Matijevic, R., Jones, S.V., Garden, A.S., Fraser, W.D., Diver, M.J., Kingsland, C.R. **Exaggerated effects of progestogen on uterine artery pulsatility index in Turner's syndrome patients receiving hormone replacement therapy.** *Fertility and Sterility* 1995; 64: 1104–1108
- Burgos, N., Cintron, D., Latortue-Albino, P., Serrano, V., Rodriguez Gutierrez, R., Faubion, S., Spencer-Bonilla, G., Erwin, P.J., Murad, M.H. **Estrogen-based hormone therapy in women with primary ovarian insufficiency: a systematic review.** *Endocrine* 2017; 58: 413–425
- Cardona Attard, C., Cameron-Pimblett, A., Puri, D., Elliot, J., Wilson, J.C., Talaulikar, V.S., Davies, M.C., Conway, G.S. **Fracture rate in women with oestrogen deficiency – comparison of Turner syndrome and premature ovarian insufficiency.** *Clinical Endocrinology* 2019; 91: 743–749
- Cartwright, B., Robinson, J., Seed, P.T., Fogelman, I., Rymer, J. **Hormone replacement therapy versus the combined oral contraceptive pill in premature ovarian failure: a randomized controlled trial of the effects on bone mineral density.** *The Journal of Clinical Endocrinology and Metabolism* 2016; 101: 3497–3505
- Castelo-Branco, C., Pons, F., Gonzalez-Merlo, J. **Bone mineral density in surgically postmenopausal women receiving hormonal replacement therapy as assessed by dual photon absorptiometry.** *Maturitas* 1993; 16: 133–137
- Castelo-Branco, C., Rovira, M., Pons, F., Durán, M., Sierra, J., Vives, A., Balasch, J., Fortuny, A., Vanrell, J. **The effect of hormone replacement therapy on bone mass in patients with ovarian failure due to bone marrow transplantation.** *Maturitas* 1996; 23: 307–312
- Chatterjee, R., Katz, M., Bajoria, R. **Use of hormone replacement therapy for correction of high turnover bone disease in hypogonadal β -thalassaemia major patients presenting with osteoporosis: comparison with idiopathic premature ovarian failure.** *Hemoglobin* 2011; 35: 653–658
- Christ, J.P., Gunning, M.N., Palla, G., Eijkemans, M.J.C., Lambalk, C.B., Laven, J.S.E., Fauser, B. **Estrogen deprivation and cardiovascular disease risk in primary ovarian insufficiency.** *Fertility and Sterility* 2018; 109: 594–600
- Coulam, C.B., Adamson, S.C., Annegers, J.F. **Incidence of premature ovarian failure.** *Obstetrics and Gynecology* 1986; 67: 604–606
- Crofton, P.M., Evans, N., Bath, L.E., Warner, P., Whitehead, T.J., Critchley, H.O., Kelnar, C.J., Wallace, W.H. **Physiological versus standard sex steroid replacement in young women with premature ovarian failure: effects on bone mass acquisition and turnover.** *Clinical Endocrinology* 2010; 73: 707–714
- Dragojević Dikić, S., Vasiljević, M., Jovanović, A., Dikić, S., Jurišić, A., Srbinović, L., Vujović, S. **Premature ovarian insufficiency – novel hormonal approaches in optimizing fertility.** *Gynecological Endocrinology* 2020; 36: 162–165
- Eeles, R.A., Tan, S., Wiltshaw, E., Fryatt, I., A'Hern, R.P., Shepherd, J.H., Harmer, C.L., Blake, P.R., Chilvers, C.E. **Hormone replacement therapy and survival after surgery for ovarian cancer.** *BMJ (Clinical Research Ed.)* 1991; 302: 259–262
- Elkind-Hirsch, K.E., Sherman, L.D., Malinak, R. **Hormone replacement therapy alters insulin sensitivity in young women with premature ovarian failure.** *The Journal of Clinical Endocrinology and Metabolism* 1993; 76: 472–475
- Elkind-Hirsch, K.E., Wallace, E., Stach, B.A., Jerger, J.F. **Cyclic steroid replacement alters auditory brainstem responses in young women with premature ovarian failure.** *Hearing Research* 1992; 64: 93–98
- Emans, S.J., Grace, E., Hoffer, F.A., Gundberg, C., Ravnkar, V., Woods, E.R. **Estrogen deficiency in adolescents and young adults: impact on bone mineral content and effects of estrogen replacement therapy.** *Obstetrics and Gynecology* 1990; 76: 585–592
- Fatemi, H.M., Bourgain, C., Donoso, P., Blockeel, C., Papanikolaou, E.G., Popovic-Todorovic, B., Devroey, P. **Effect of oral administration of dydrogesterone versus vaginal administration of natural micronized progesterone on the secretory transformation of endometrium and luteal endocrine profile in patients with premature ovarian failure: a proof of concept.** *Human Reproduction (Oxford, England)* 2007; 22: 1260–1263
- Gazarra, L.B.C., Bonacordi, C.L., Yela, D.A., Benetti-Pinto, C.L. **Bone mass in women with premature ovarian insufficiency: a comparative study between hormone therapy and combined oral contraceptives.** *Menopause (New York, N.Y.)* 2020; 27: 1110–1116
- Giraldo, H., Benetti-Pinto, C., Ferreira, V., Garmes, H., Yela, D., Giraldo, P. **Standard hormone therapy is inadequate for bone density in premature ovarian insufficiency.** *Gynecological Endocrinology* 2017; 33: 283–286
- Gokmen, O., Yapar Eyi, E.G. **Hormone replacement therapy and lipid-lipoprotein concentrations.** *European Journal of Obstetrics, Gynecology, and Reproductive Biology* 1999; 85: 31–41
- Guerrieri, G.M., Martinez, P.E., Klug, S.P., Haq, N.A., Vanderhoof, V.H., Koziol, D.E., Papat, V.B., Kalantaridou, S.N., Calis, K.A., Rubinow, D.R., Schmidt, P.J., Nelson, L.M. **Effects of physiologic testosterone therapy on quality of life, self-esteem, and mood in women with primary ovarian insufficiency.** *Menopause (New York, N.Y.)* 2014; 21: 952–961
- Guyatt, G., Oxman, A.D., Akl, E.A., Kunz, R., Vist, G., Brozek, J., Norris, S., Falck-Ytter, Y., Glasziou, P., DeBeer, H., Jaeschke, R., Rind, D., Meerpohl, J., Dahm, P., Schunemann, H.J. **GRADE guidelines: 1. Introduction-GRADE evidence profiles and summary of findings tables.** *J. Clin. Epidemiol.* 2011; 64: 383–394
- Haines, C.J., Chung, T.K., Masarei, J.R., Tomlinson, B., Lau, J.T. **The effect of percutaneous oestrogen replacement therapy on Lp(a) and other lipoproteins.** *Maturitas* 1995; 22: 219–225
- Hartmann, B.W., Kirchengast, S., Albrecht, A., Huber, J.C., Söregi, G. **Effect of hormone replacement therapy on growth hormone stimulation in women with premature ovarian failure.** *Fertility and Sterility* 1997; 68: 103–107
- Higgins, J.P.T., Green, S. 2011 *Cochrane Handbook for Systematic Reviews of Intervention The Cochrane Collaboration*
- Honigberg, M.C., Zekavat, S.M., Aragam, K., Finneran, P., Klarin, D., Bhatt, D.L., Januzzi, J.L. Jr., Scott, N.S., Natarajan, P. **Association of premature natural and surgical menopause with incident cardiovascular disease.** *JAMA* 2019; 322: 2411–2421
- Jungar, S.B., Chauhan, B.G. **Prevalence and determinants of premature menopause among Indian women: issues and challenges ahead.** *Health and Social Work* 2017; 42: 79–86
- Kalantaridou, S.N., Naka, K.K., Papanikolaou, E., Kazakos, N., Kravariti, M., Calis, K.A., Paraskevaidis, E.A., Sideris, D.A., Tsatsoulis, A., Chrousos, G.P., Michalis, L.K. **Impaired endothelial function in young women with premature ovarian failure: normalization with hormone therapy.** *The Journal of Clinical*

- Endocrinology and Metabolism 2004; 89: 3907–3913
- Kapetanakis, E., Dmowski, W.P., Auletta, F., Scommegna, A. **Endocrine and clinical effects of estradiol and testosterone pellets used in long-term replacement therapy.** International Journal of Gynaecology and Obstetrics 1982; 20: 387–399
- Khastgir, G., Studd, J.W., Fox, S.W., Jones, J., Alagband-Zadeh, J., Chow, J.W. **A longitudinal study of the effect of subcutaneous estrogen replacement on bone in young women with Turner's syndrome.** Journal of Bone and Mineral Research 2003; 18: 925–932
- Kilic, S., Yilmaz, N., Erdogan, G., Aydin, M., Tasdemir, N., Doganay, M., Batioglu, S. **Effect of non-oral estrogen on risk markers for metabolic syndrome in early surgically menopausal women.** Climacteric 2010; 13: 55–62
- Kim, H.J., Lee, D.-Y., Yoon, B.-K., Choi, D. **Uterine development after estrogen replacement therapy in women with different etiologies of primary hypogonadism.** Journal of Pediatric and Adolescent Gynecology 2016; 29: 344–347
- Kraus, J., Unanue, N., Espinoza, A., Iniguez, G., Torres-Santiago, L., Singh, R., Mauras, N., Mericq, V. **Is there a difference between ultrasonographic (US) uterine changes of oral versus transdermal (TD) 17beta estradiol (17beta E2) in girls with Turner syndrome (TS)? Own experience and literature review.** Pediatric Endocrinology Reviews 2018; 16: 178–185
- Kung, A.W., Chan, T.M., Lau, C.S., Wong, R.W., Yeung, S.S. **Osteopenia in young hypogonadal women with systemic lupus erythematosus receiving chronic steroid therapy: a randomized controlled trial comparing calcitriol and hormonal replacement therapy.** Rheumatology (Oxford, England) 1999; 38: 1239–1244
- Kurabayashi, T., Yasuda, M., Fujimaki, T., Yamamoto, Y., Oda, K., Tanaka, K. **Effect of hormone replacement therapy on spinal bone mineral density and T lymphocyte subsets in premature ovarian failure and Turner's syndrome.** International Journal of Gynaecology and Obstetrics 1993; 42: 25–31
- La Marca, A., Mastellari, E. **Fertility preservation for genetic diseases leading to premature ovarian insufficiency (POI).** Journal of Assisted Reproduction and Genetics 2021; 38: 759–777
- Lambrinoudaki, I., Paschou, S.A., Lumsden, M.A., Faubion, S., Makrakis, E., Kalantaridou, S., Panay, N. **Premature ovarian insufficiency: a toolkit for the primary care physician.** Maturitas 2021; 147: 53–63
- Langrish, J.P., Mills, N.L., Bath, L.E., Warner, P., Webb, D.J., Kelnar, C.J., Critchley, H.O., Newby, D.E., Wallace, W.H. **Cardiovascular effects of physiological and standard sex steroid replacement regimens in premature ovarian failure.** Hypertension (Dallas, Tex.: 1979) 2009; 53: 805–811
- Li, T.C., Cooke, I.D., Warren, M.A., Goolamallee, M., Graham, R.A., Aplin, J.D. **Endometrial responses in artificial cycles: a prospective study comparing four different oestrogen dosages.** British Journal of Obstetrics and Gynaecology 1992; 99: 751–756
- Li, T.C., Dockery, P., Ramsewak, S.S., Klentzeris, L., Lenton, E.A., Cooke, I.D. **The variation of endometrial response to a standard hormone replacement therapy in women with premature ovarian failure. An ultrasonographic and histological study.** British Journal of Obstetrics and Gynaecology 1991; 98: 656–661
- Lip, G.Y., Blann, A.D., Jones, A.F., Beevers, D.G. **Effects of hormone-replacement therapy on hemostatic factors, lipid factors, and endothelial function in women undergoing surgical menopause: implications for prevention of atherosclerosis.** American Heart Journal 1997; 134: 764–771
- Luborsky, J.L., Meyer, P., Sowers, M.F., Gold, E.B., Santoro, N. **Premature menopause in a multi-ethnic population study of the menopause transition.** Human Reproduction 2003; 18: 199–206
- Lyrithis, G.P., Karpathios, S., Basdekis, K., Grigoriou, O., Katostaras, T., Paspatis, I., Stamatopoulos, T., Siampalioti, G., Lyrithis, P.G. **Prevention of post-oophorectomy bone loss with tibolone.** Maturitas 1995; 22: 247–253
- Madalinska, J.B., van Beurden, M., Bleiker, E.M., Valdimarsdottir, H.B., Hollenstein, J., Massuger, L.F., Gaarenstroom, K.N., Mourits, M.J., Verheijen, R.H., van Dorst, E.B., van der Putten, H., van der Velden, K., Boonstra, H., Aaronson, N.K. **The impact of hormone replacement therapy on menopausal symptoms in younger high-risk women after prophylactic salpingo-oophorectomy.** Journal of Clinical Oncology 2006; 24: 3576–3582
- Mishra, G.D., Pandeya, N., Dobson, A.J., Chung, H.F., Anderson, D., Kuh, D., Sandin, S., Giles, G.G., Bruinsma, F., Hayashi, K., Lee, J.S., Mizunuma, H., Cade, J.E., Burley, V., Greenwood, D.C., Goodman, A., Simonsen, M.K., Adami, H.O., Demakakos, P., Weiderpass, E. **Early menarche, nulliparity and the risk for premature and early natural menopause.** Human Reproduction (Oxford, England) 2017; 32: 679–686
- Mittal, M., Savvas, M., Arya, R., McEniery, C., Narvekar, N., Cardozo, L., Panay, N., Hamoda, H. **A randomised controlled trial comparing the effects of micronized progesterone to medroxyprogesterone acetate on cardiovascular health, lipid metabolism and the coagulation cascade in women with premature ovarian insufficiency: study protocol and review of the literature.** Menopause International 2013; 19: 127–132
- Modesti, P.A., Reboldi, G., Cappuccio, F.P., Agemam, C., Remuzzi, G., Rapi, S., Perruolo, E., Parati, G. **Panethnic differences in blood pressure in Europe: a systematic review and meta-analysis.** PloS One 2016; 11:e0147601
- Naessén, S., Bergström, I., Ljungman, P., Landgren, B.-M. **Long-term follow-up of bone density, general and reproductive health in female survivors after treatment for haematological malignancies.** European Journal of Haematology 2014; 93: 137–142
- Nakamura, T., Tsuburai, T., Tokinaga, A., Nakajima, I., Kitayama, R., Imai, Y., Nagata, T., Yoshida, H., Hirahara, F., Sakakibara, H. **Efficacy of estrogen replacement therapy (ERT) on uterine growth and acquisition of bone mass in patients with Turner syndrome.** Endocrine Journal 2015; 62: 965–970
- O'Donnell, R.L., Warner, P., Lee, R.J., Walker, J., Bath, L.E., Kelnar, C.J., Wallace, W.H., Critchley, H.O. **Physiological sex steroid replacement in premature ovarian failure: randomized crossover trial of effect on uterine volume, endometrial thickness and blood flow, compared with a standard regimen.** Human Reproduction (Oxford, England) 2012; 27: 1130–1138
- Page, M.J., McKenzie, J.E., Bossuyt, P.M., Boutron, I., Hoffmann, T.C., Mulrow, C.D., Shamseer, L., Tetzlaff, J.M., Akl, E.A., Brennan, S.E., Chou, R., Glanville, J., Grimshaw, J.M., Hróbjartsson, A., Lalu, M.M., Li, T., Loder, E.W., Mayo-Wilson, E., McDonald, S., McGuinness, L.A., Stewart, L.A., Thomas, J., Tricco, A.C., Welch, V.A., Whiting, P., Moher, D. **The PRISMA 2020 statement: an updated guideline for reporting systematic reviews.** BMJ 2021; 372: n71
- Panay, N., Anderson, R.A., Nappi, R.E., Vincent, A.J., Vujovic, S., Webber, L., Wolfman, W. **Premature ovarian insufficiency: an International Menopause Society White Paper.** Climacteric 2020; 23: 426–446
- Popat, V.B., Calis, K.A., Kalantaridou, S.N., Vanderhoof, V.H., Koziol, D., Troendle, J.F., Reynolds, J.C., Nelson, L.M. **Bone mineral density in young women with primary ovarian insufficiency: results of a three-year randomized controlled trial of physiological transdermal estradiol and testosterone replacement.** The Journal of Clinical Endocrinology and Metabolism 2014; 99: 3418–3426
- Progetto Menopausa Italia Study Group. **Premature ovarian failure: frequency and risk factors among women attending a network of menopause clinics in Italy.** BJOG 2003; 110: 59–63
- Review Manager (RevMan) [Computer program] Version 5.4. The Cochrane Collaboration, 2020.
- Ross, J.L., Quigley, C.A., Cao, D., Feuillan, P., Kowal, K., Chipman, J.J., Cutler, G.B. Jr. **Growth hormone plus childhood low-dose estrogen in Turner's syndrome.** New England Journal of Medicine 2011; 364: 1230–1242
- Shah, S., Forghani, N., Durham, E., Neely, E.K. **A randomized trial of transdermal and oral estrogen therapy in adolescent girls with hypogonadism.** International Journal of Pediatric Endocrinology 2014; 2014: 12
- Shea, K.L., Gavin, K.M., Melanson, E.L., Gibbons, E., Stavros, A., Wolfe, P., Kittelson, J.M., Vondracek, S.F., Schwartz, R.S., Wierman, M.E., Kohrt, W.M. **Body composition and bone mineral density after ovarian hormone suppression with or without estradiol treatment.** Menopause (New York, N.Y.) 2015; 22: 1045–1052
- Sherwin, B.B., Gelfand, M.M. **Differential symptom response to parenteral estrogen and/or androgen administration in the surgical menopause.** American Journal of Obstetrics and Gynecology 1985; 151: 153–160
- Sherwin, B.B., Gelfand, M.M. **Sex steroids and affect in the surgical menopause: a double-blind, crossover study.** Psychoneuroendocrinology 1985; 10: 325–335
- Somunkiran, A., Erel, C.T., Demirci, F., Senturk, M.L. **The effect of tibolone versus 17beta-estradiol on climacteric symptoms in women with surgical menopause: a randomized, crossover study.** Maturitas 2007; 56: 61–68
- Vermeulen, R.F.M., Beurden, M.V., Kieffer, J.M., Bleiker, E.M.A., Valdimarsdottir, H.B., Massuger, L., Mourits, M.J.E., Gaarenstroom, K.N., van Dorst, E.B.L., van der Putten, H., Aaronson, N.K. **Hormone replacement therapy after risk-reducing salpingo-oophorectomy minimises endocrine and sexual problems: a prospective**

- study.** *European Journal of Cancer* (Oxford, England: 1990) 2017; 84: 159–167
- Webber, L., Anderson, R.A., Davies, M., Janse, F., Vermeulen, N. **HRT for women with premature ovarian insufficiency: a comprehensive review.** *Human Reproduction Open* 2017; 2017: hox007
- Webber, L., Davies, M., Anderson, R., Bartlett, J., Braat, D., Cartwright, B., Cifkova, R., de Muinck Keizer-Schrama, S., Hogervorst, E., Janse, F., Liao, L., Vlasisavljevic, V., Zillikens, C., Vermeulen, N. **ESHRE guideline: management of women with premature ovarian insufficiency.** *Human Reproduction* (Oxford, England) 2016; 31: 926–937
- Wells, G.A., Shea, B., O'Connell, D., Peterson, J., Welch, V., Losos, M., Tugwell, P., 2000. The Newcastle-Ottawa Scale (NOS) for assessing the quality of nonrandomised studies in meta-analyses. Available at http://www.ohri.ca/programs/clinical_epidemiology/oxford.asp
- Wong, Q.H.Y., Yeung, T.W.Y., Yung, S.S.F., Ko, J.K.Y., Li, H.W.R., Ng, E.H.Y. **The effect of 12-month dehydroepiandrosterone supplementation on the menstrual pattern, ovarian reserve markers, and safety profile in women with premature ovarian insufficiency.** *Journal of Assisted Reproduction and Genetics* 2018; 35: 857–862
- Yang, X., Wang, C., He, X., Wei, J., Wang, Y., Li, X., Xu, L.P. **Hormone therapy for premature ovarian insufficiency patients with malignant hematologic diseases.** *Climacteric* 2017; 20: 268–273
- Zuckerman-Levin, N., Frolova-Bishara, T., Militianu, D., Levin, M., Aharon-Peretz, J., Hochberg, Z. **Androgen replacement therapy in Turner syndrome: a pilot study.** *The Journal of Clinical Endocrinology and Metabolism* 2009; 94: 4820–4827

Received 2 November 2021; received in revised form 12 January 2022; accepted 8 February 2022.